Multi-Scale Spectral Kernel Machines for Tabular Data*

Agus Sudjianto Aijun Zhang

June 12, 2026

Abstract

We treat supervised learning as the construction of a kernel: once a positive-semidefinite kernel is fixed, kernel ridge regression determines the estimator, so learning is the choice of a data-adapted kernel. We develop the Multi-Scale Spectral Kernel Machine (MS-SKM), a structured, inspectable kernel for tabular data, and study it as a ladder in which each rung lifts one modeling restriction: from an additive model on fixed random Fourier features, to learned frequencies, to a radial kernel on a per-coordinate spectral embedding, to feature mixing, to multi-resolution banks fused on the simplex. Three results organize the design. Interactions come from the kernel nonlinearity, not the linear mixer, which only rotates them from axis-aligned to oblique. The convex multi-bank fuse is positive-semidefinite and decomposes into one component per bank, whereas a linear mixer over banks collapses to a single bank with the pooled frequencies—so the per-bank bandwidth is what fusion adds. A parameter accounting makes the per-band frequency count the overfitting axis and the bank count the multi-scale axis. The ladder isolates each effect on real data, and a variational head trains the same kernel by a sparse Gaussian-process ELBO to return a calibrated predictive distribution.

Keywords: kernel methods, spectral kernels, Gaussian processes, tabular data, kernel ridge regression, additive models, multi-resolution learning, variational inference

1 Introduction

A prediction at a query point borrows strength from training points through a notion of similarity, the *kernel*. Kernel methods fix this object a priori, tree ensembles induce it through a learned partition, and neural networks learn a representation whose inner product is a kernel. Gradient-boosted decision trees remain the default for tabular data and frequently outperform deep models [3, 8, 11, 18]. Taking the kernel itself as the primitive, we ask what a trainable kernel for tabular data should look like, and we build it from the spectral domain as a competitive learned-kernel alternative.

Learnable spectral kernels are not new: spectral-mixture kernels model a spectral density [25], sparse-spectrum Gaussian processes learn finite spectral atoms [13] and à-la-carte kernels learn groups of fast frequencies [27]. Our contribution is an *arrangement* of these decisions into a tabular inductive bias and a *ladder* that makes each decision's contribution measurable. Each rung lifts one restriction:

1. **Rung 0 — fixed-frequency additive model.** Per-coordinate random Fourier features with a linear readout. The frequencies are fixed, the model is additive, there is no kernel. This is a generalized additive model [9] solved in closed form.

*Code available at `https://github.com/asudjianto-xml/Spectral-Kernel`.

2. **Rung A — learned frequencies, still additive.** The frequencies are learned. The model stays additive and kernel-free.
3. **Rung B — the kernel, no mixing.** A radial base kernel on a per-feature spectral embedding, with a block-diagonal encoder so no feature is mixed into another.
4. **Rung 1 — full mixing.** The encoder mixes features. The kernel now sees oblique interactions.
5. **Rung 2 — multi-resolution banks.** Frequencies are partitioned into H banks, each with its own bandwidth, convex-fused on the simplex.

The ladder is the methodological point. A single accuracy number hides whether a model needs interactions at all, whether they are axis-aligned or oblique and whether it needs multiple scales or only more frequencies. Lifting one restriction at a time answers each question separately, on the same data and the same training machinery.

Contributions. We are explicit about what is *proven*, what is a *diagnostic*, and what is a *conjecture supported by a sweep.*

1. *(Proven.)* The per-coordinate feature map realizes a sum of one-dimensional spectral-mixture kernels (Prop. 1), dense in the additive-stationary class between additive Gaussian processes [4] and spectral-mixture kernels [25].
2. *(Proven / diagnostic.)* Interactions come from the radial base kernel, not the mixer (Prop. 2). A linear mixer is a learned metric $W^\top W$ whose off-diagonal blocks are an inspectable diagnostic of *metric* interaction; with the mixer block-diagonal the kernel is still non-additive.
3. *(Proven.)* The convex multi-bank fuse is positive-semidefinite and decomposes exactly into one component per bank in the sum RKHS (Prop. 4), and a single linear mixer over banks collapses to one bank with the pooled frequencies (Prop. 5)—so the per-bank bandwidth, not the bank count, is what convex fusion adds.
4. *(Accounting + sweep.)* The dominant parameter count scales with the per-band frequency count K and is independent of the bank count H (Prop. 6); we observe K to be the overfitting axis and H the multi-scale axis.
5. *(Whittle / diagnostic.)* A one-dimensional criterion for when learning the frequencies helps (Prop. 7): the gradient on a frequency atom is proportional to the local residual spectral mass, vanishing on flat spectra and active on peaked ones.
6. *(Predictive distribution.)* A variational head (Sec. 8) that trains the same learned kernel by the ELBO of a sparse Gaussian process [10, 24], giving a calibrated Gaussian predictive distribution rather than a point.
7. *(Interpretability.)* Because the kernel is learned, its parameters are the explanation: automatic relevance determination (ARD) gives a per-feature importance and the encoder metric $W^\top W$ gives pairwise metric interactions, read directly off the fitted model with no surrogate (Sec. 9).
8. An open-source implementation and an ablation ladder that isolates each effect on real tabular data.

2 The kernel is the object of learning

Let $\{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathcal{X} \subseteq \mathbb{R}^d$. A symmetric positive-semidefinite (PSD) kernel k induces a reproducing-kernel Hilbert space (RKHS) $\mathcal{H}_k$ and a Gaussian-process prior $f \sim \mathcal{GP}(0, k)$. Regularized risk minimization in $\mathcal{H}_k$ has, by the representer theorem, the finite solution $\hat{f}(\cdot) = \sum_i \alpha_i k(\cdot, x_i)$; for squared loss this is kernel ridge regression (KRR), which coincides with the GP posterior mean

$$\hat{f}(x) = k(x, X)\,(K + \lambda I)^{-1} y, \qquad K_{ij} = k(x_i, x_j). \tag{1}$$

The estimator (1) is a fixed functional of k; the learning problem is the choice of k from a parameterized family $\{k_\theta\}$, here by empirical Bayes over θ [20]. We therefore study the admissible k_θ directly.

3 A per-coordinate spectral construction

A continuous stationary kernel $k(x, x') = \kappa(x - x')$ is PSD if and only if κ is the Fourier transform of a finite nonnegative measure μ, the spectral measure (Bochner). Kernel learning restricted to the stationary class is spectral-measure estimation [25]. Rather than estimate μ on $\mathbb{R}^d$ jointly, MS-SKM estimates a measure on each coordinate. Fix per-coordinate frequencies $\{\omega_{j,k}\}_{k=1}^K$, nonnegative amplitudes $\{a_{j,k}\}$ and a per-coordinate automatic-relevance-determination (ARD) scale $s_j > 0$ [14, 16, 20], and define the feature map with components

$$\psi(x)_{j,k} = a_{j,k}\big[\cos(2\pi s_j\, \omega_{j,k}\, x_j),\ \sin(2\pi s_j\, \omega_{j,k}\, x_j)\big], \qquad j = 1, \dots, d,\ k = 1, \dots, K. \tag{2}$$

Unlike random Fourier features [19], where each feature is $\cos(\omega^\top x)$ with $\omega \in \mathbb{R}^d$ mixing all coordinates, (2) is a one-dimensional Fourier expansion applied independently to each coordinate—the periodic numerical embedding of tabular deep learning [7, 28], read spectrally.

Proposition 1 (Additive spectral-mixture realization)**.** *The inner product of* (2) *is the additive kernel*

$$\langle \psi(x), \psi(x') \rangle = \sum_{j=1}^{d} \sum_{k=1}^{K} a_{j,k}^2\, \cos\big(2\pi s_j \omega_{j,k}(x_j - x_j')\big) = \sum_{j=1}^{d} \kappa_j(x_j - x_j'), \tag{3}$$

where each κ_j is a one-dimensional stationary kernel with atomic spectral measure

$$\mu_j = \tfrac{1}{2} \sum_k a_{j,k}^2 \big(\delta_{s_j \omega_{j,k}} + \delta_{-s_j \omega_{j,k}}\big).$$

Hence ψ realizes a sum of per-coordinate spectral-mixture kernels and is PSD. As $K \to \infty$ with $\{a_{j,k}^2\}$ a consistent quadrature of a density p_j, each κ_j converges to the kernel of p_j, so the family is dense in the additive-stationary class [4, 25].

Proof. The identity $\cos A \cos B + \sin A \sin B = \cos(A - B)$ with $A = 2\pi s_j \omega_{j,k} x_j$ and $B = 2\pi s_j \omega_{j,k} x_j'$ gives the cosine term after the $a_{j,k}^2$ weighting; summing over k then j yields (3). Each μ_j is finite, nonnegative and symmetric, so by Bochner κ_j is PSD; a sum of PSD kernels in disjoint coordinates is PSD. Density is coordinatewise [25] and additive [4]. □

With a linear readout $f(x) = \beta^\top \psi(x)$ this is rung 0 (frequencies fixed, solved in closed form) and rung A (frequencies learned by gradient descent). In both, the readout weights are the per-frequency amplitudes, the model is additive, and there is no kernel.

4 Interactions from the kernel, not the mixer

The additive map (3) has no cross-coordinate interactions. We introduce a single linear encoder W, $\phi(x) = W\psi(x)$, and a Laplace base kernel,

$$k_\theta(x, x') = \exp\big(-\|W(\psi(x) - \psi(x'))\|_2/T\big), \qquad \theta = (s, \omega, a, W, T), \tag{4}$$

the deep-kernel construction [26] with a shallow, linear warp on additive spectral features. It is essential to separate three notions of interaction. *Metric:* cross-coordinate blocks of $M = W^\top W$. *Kernel:* failure of k_θ to be additive. *Functional:* a non-additive contribution to the fitted predictor.

Proposition 2 (Where interactions come from)**.** *Let $\Delta\psi = \psi(x) - \psi(x')$, blocked by coordinate. Then k_θ in* (4) *is PSD and*

$$\|W\Delta\psi\|_2^2 = \Delta\psi^\top M \Delta\psi = \sum_j \Delta\psi^{(j)\top} M_{jj} \Delta\psi^{(j)} + \sum_{j \neq j'} \Delta\psi^{(j)\top} M_{jj'} \Delta\psi^{(j')}, \quad M = W^\top W \succeq 0. \tag{5}$$

Consequently (i) *a linear readout on ψ (no kernel) is additive for any W, since $\beta^\top W\psi$ is linear in ψ and each block of ψ depends on one coordinate, so a linear mixer neither creates nor removes interactions;* (ii) *with M block-diagonal the kernel is still non-additive, because* $\exp(-\sqrt{\sum_j d_j^2}/T)$ *does not factor over the d_j, so kernel and functional interactions are present with zero metric interaction;* (iii) *the off-diagonal blocks $M_{jj'}$ are an inspectable diagnostic of* metric *interaction, and mixing rotates the interactions the kernel forms from axis-aligned to oblique.*

Proof. $\exp(-\|Lz\|/T)$ is PSD for any matrix L and $T > 0$: $\exp(-\|u\|/T)$ is PSD (Matérn-$\frac{1}{2}$) and composition with a linear map preserves positive-definiteness; take $L = W$. The decomposition is the block expansion of $\Delta\psi^\top W^\top W \Delta\psi$. Claim (i) is immediate from linearity. For (ii), with M block-diagonal the squared distance is $\sum_j d_j^2$ but the Laplace map of its square root does not separate. □

Proposition 3 (Universality)**.** *If the per-coordinate bank separates points and $\phi = W\psi$ is injective on the compact domain, then k_θ is universal: its RKHS is dense in $C(\mathcal{X})$ [15]. Every continuous function, and thus every interaction, is approximable; the additive base is not a structural barrier. Interactions enter through the radial nonlinearity acting on linear projections of the univariate features, exactly as an RBF kernel on raw inputs realizes interactions despite a coordinatewise input.*

Rung B is (4) with W block-diagonal (a per-feature encoder, no mixing); rung 1 is (4) with W full. By Prop. 2 rung B already has interactions, axis-aligned; rung 1 makes them oblique.

5 Multi-resolution fusion

We replicate the construction in H banks. Bank h carries its own frequencies ω_h and amplitudes a_h initialized on a band of a log-spaced partition of the frequency range, its own bandwidth T_h, and a shared encoder W applied to each bank's own $2dK$-dimensional feature vector $\psi_h(x)$ separately. The banks are fused convexly,

$$k(x, x') = \sum_{h=1}^{H} w_h \, k_h(x, x'), \qquad w = \mathrm{softmax}(\cdot) \in \Delta^{H-1}, \tag{6}$$

with k_h as in (4) using (ω_h, a_h, T_h). This is rung 2.

Proposition 4 (Closure and decomposition)**.** *For $w_h \geq 0$, k in (6) is PSD. The fused KRR fit decomposes exactly into one component per bank,*

$$\hat{g}_h = w_h K_h (K + \lambda I)^{-1} y, \qquad \textstyle\sum_h \hat{g}_h = \hat{f}, \tag{7}$$

and, by Aronszajn's sum-of-kernels theorem [1, 2], the RKHS of $k = \sum_h w_h k_h$ is the sum space $\sum_h \mathcal{H}_{k_h}$ with the infimal-convolution norm $\|f\|^2_{\mathcal{H}_k} = \min_{f=\sum_h g_h} \sum_h \frac{1}{w_h} \|g_h\|^2_{\mathcal{H}_{k_h}}$. If each bank is normalized ($k_h(x,x) = 1$) the posterior mean is invariant along $(w, \lambda) \mapsto (\alpha w, \alpha\lambda)$, so the simplex fixes the mixture scale.

Proof. A nonnegative combination of PSD matrices is PSD. Equation (7) is Gaussian conditioning: with independent priors $g_h \sim \mathcal{GP}(0, w_h k_h)$ and $\mathrm{Cov}(g_h, y) = w_h K_h$, $\mathbb{E}[g_h \mid y] = w_h K_h (K + \lambda I)^{-1} y$, and $\sum_h w_h K_h = K$. The sum-RKHS norm is Aronszajn's theorem. Invariance follows from $\alpha K(\alpha K + \alpha\lambda I)^{-1} = K(K + \lambda I)^{-1}$. □

The decomposition is the rigorous statement that a convex fuse of banks at different scales is *not* reducible to a single kernel. The next proposition shows what the alternative — mixing the banks linearly — actually is.

Proposition 5 (A linear mixer over banks collapses to one bank)**.** *Let the banks be combined by a single linear map applied to the concatenation $[\psi_1(x); \ldots; \psi_H(x)]$, followed by one base kernel of bandwidth T. The concatenation of H banks of K frequencies is a single bank of HK frequencies, and one linear encoder over it is the encoder of (4) with $K' = HK$. Hence "H banks + one linear mixer" equals "one bank with $K' = HK$ frequencies": a single kernel with a single bandwidth. The per-bank bandwidths T_h of (6) are absent from this construction, and a single bandwidth cannot span frequencies of widely separated scale.*

Proposition 6 (Parameter accounting)**.** *With H banks of K frequencies each and the shared per-bank encoder $W \in \mathbb{R}^{d_\phi \times 2dK}$, the free-parameter count is*

$$\underbrace{H \cdot d \cdot 2K}_{\textit{bank spectra}} + \underbrace{d \cdot 2K \cdot d_\phi}_{\textit{shared encoder } W} + O(H), \tag{8}$$

so the dominant encoder term scales with K and is independent of H.

Remark 1 (Banks versus frequencies)**.** The two axes generalize differently. Increasing H adds multi-resolution structure on a shared basis, gated by the simplex weights w_h. Increasing K widens the dominant ungated encoder (8) and packs more frequencies into a band whose amplitude envelope is already a smooth quadrature (Prop. 1). We therefore expect K to be the overfitting axis and H the multi-scale axis, a multi-resolution echo of the frequency-overfitting of [5]. The selection of w here is by marginal likelihood, jointly with the banks; selecting w by an unbiased risk estimate over fixed banks [6, 23] would inherit the oracle guarantee that convex fusion is never worse than the best single bank, and is a natural alternative.

6 When does learning the frequencies help?

Sparse-spectrum methods learn ω by marginal likelihood and overfit on smooth targets [5, 13]. We give a partial criterion in the cleanest setting, one-dimensional regression, and use it as a tabular diagnostic.

Proposition 7 (Frequency-learning gradient, 1-D Whittle regime)**.** *Under the Whittle approximation to the Gaussian log-likelihood, with model spectrum $S_\theta(\omega) = \lambda + \sum_k a_k^2 g(\omega - \omega_k)$, the gradient moving an atom is*

$$\frac{\partial \mathcal{L}}{\partial \omega_k} \;\propto\; a_k^2\, \partial_\omega r(\omega)\big|_{\omega_k}, \qquad r(\omega) = \hat{p}(\omega) - \textstyle\sum_{k'} a_{k'}^2 g(\omega - \omega_{k'}), \tag{9}$$

the residual spectral density ($\hat{p}$ the periodogram, g a fixed atom kernel). If $\hat{p}$ is locally flat the atoms do not move and a fixed grid is already a consistent quadrature; if $\hat{p}$ is peaked the atoms migrate to the lines and a learned grid strictly improves the fit.

Sketch. Differentiate $-\frac{1}{2}\int[\log S_\theta + \hat{p}/S_\theta]d\omega$ in ω_k using $\partial S_\theta/\partial\omega_k = -a_k^2 g'(\omega - \omega_k)$; the leading term is $a_k^2 \partial_\omega r(\omega_k)$. □

For multivariate tabular data the per-coordinate spectral density is not canonically defined, so we read this as a diagnostic: learning the frequencies helps in proportion to per-coordinate spectral peakedness. This is consistent with the smooth-versus-periodic reversal in Section 9.

7 Training and inference

Hyperparameters θ are fit by the negative log marginal likelihood $\mathcal{L} = \frac{1}{2} y^\top (K + \lambda I)^{-1} y + \frac{1}{2}\log|K + \lambda I| + \text{const}$, optimized by gradient descent on random support subsets (Cholesky in double precision, autograd); the noise λ is the learned ridge. Prediction is full-train KRR (1), solved by a truncated Lanczos tridiagonalization [12] so the cost is a handful of matrix-vector products rather than a dense factorization. The frequencies and amplitudes are learned, never frozen, so the construction is not random Fourier features: we learn the kernel rather than approximate a fixed one. The feature and label scalers are fit on train, the ridge and the calibration temperature on a validation fold, and test data is touched only at prediction.

8 Predictive distributions via a variational bound

The NLML objective with the KRR decode returns the Gaussian-process posterior mean — a point prediction. The exact GP posterior also has a closed-form predictive variance, but it requires a dense solve and is tied to the Gaussian likelihood. To obtain a predictive distribution that scales by minibatching and extends to non-Gaussian likelihoods, we train the same learned kernel by the evidence lower bound (ELBO) of a sparse variational Gaussian process [10, 24].

Place M inducing inputs Z with prior $p(u) = \mathcal{N}(0, K_{ZZ})$, and a whitened variational posterior $q(u) = \mathcal{N}(Lm,\, LSL^\top)$ with $L = \mathrm{chol}(K_{ZZ})$. The variational marginal at a point is $q(f_i) = \mathcal{N}(\mu_i, s_i^2)$ with

$$\mu_i = b_i^\top m, \qquad s_i^2 = k_{ii} - b_i^\top (I - S)\, b_i, \qquad b_i = L^{-1} k(Z, x_i), \tag{10}$$

and the kernel's unit diagonal gives $k_{ii} = 1$. For a Gaussian likelihood with noise σ^2 the bound is

$$\mathrm{ELBO} = \sum_i \Big[\log \mathcal{N}(y_i \mid \mu_i, \sigma^2) - \tfrac{s_i^2}{2\sigma^2}\Big] - \mathrm{KL}\big(q(u)\,\|\,p(u)\big), \quad \mathrm{KL} = \tfrac{1}{2}\big[\,\mathrm{tr}\, S + \|m\|^2 - M - \log|S|\big]. \tag{11}$$

The spectral parameters, the inducing inputs Z and the variational parameters (m, S, σ^2) are learned jointly by maximizing the ELBO on minibatches. The predictive distribution at $x^\star$ is Gaussian, $\mathcal{N}(\mu^\star,\, s^{\star 2} + \sigma^2)$, so every prediction carries a variance.

Table 1: `make_moons` test metrics. AUC and F1 higher is better; Brier and log loss lower is better. AUC saturates; the kernel rungs lead on F1 and Brier, and CatBoost is the least calibrated (highest Brier and log loss) despite a comparable AUC.

model	AUC	F1	Brier	LogLoss
rung 0: SpectralGAM	0.9895	0.9385	0.0430	0.1386
rung A: LearnedGAM	0.9907	0.9457	0.0430	0.1537
rung B: MSSKM(mix=False)	0.9888	0.9605	0.0358	0.1444
rung 1: MSSKM()	0.9889	0.9605	0.0350	0.1393
rung 2: MSSKM(H=4)	0.9896	0.9605	0.0367	0.1423
CatBoost	0.9893	0.9444	0.0467	0.2011

The intervals are calibrated. On California Housing the 90% and 95% central predictive intervals cover 0.93 and 0.95 of held-out points; on a one-dimensional series with a held-out gap the predictive standard deviation grows from in-sample through the gap to extrapolation, the expected Gaussian-process behavior. The sparse M-point head reaches $R^2 \approx 0.84$ where the exact decode reaches 0.88. The same bound extends to a Bernoulli or softmax likelihood for classification.

9 Experiments

We use California Housing (regression, n=20,640, d=8) and Breast Cancer (binary, n=569, d=30) from scikit-learn, and an hourly bike-sharing series (regression, periodic), each with a held-out 20% test split; the remaining data is split into train and validation for early stopping and the ridge. Classification uses one-hot KRR with a temperature-scaled posterior. The kernel is Laplace, $K = 16$ frequencies per feature per bank unless swept, $d_\phi = 128$.

An illustrative 2-D example. Figure 1 fits the ladder and a tuned CatBoost on `make_moons`, two interleaving crescents whose separation needs an oblique interaction. The additive rungs (0, A) draw blocky, near-separable boundaries; the boundary curves the moment the kernel appears (rung B), feature mixing (rung 1) wraps it obliquely around both crescents and CatBoost draws an axis-aligned staircase. Accuracy sits at the noise floor, so the metrics (Table 1) read through the boundary: AUC saturates, the kernel rungs win on F1 and Brier, and CatBoost — comparable on AUC — has markedly worse Brier and log loss, the signature of an accurate ranker with overconfident probabilities. The variational head (Sec. 8) turns the same kernel into a predictive distribution (Figure 2): the predictive standard deviation is low over the data-covered crescents and rises in the empty regions, the epistemic uncertainty of a Gaussian process.

The ladder decomposition. Table 2 reports each rung. On California the gain over the additive model is almost entirely the kernel and almost entirely axis-aligned: rung B (kernel, no mixing) recovers $0.711 \to 0.873$ of the $0.711 \to 0.876$ total, and the oblique mixing of rung 1 adds 0.003. On Breast Cancer the reverse holds: the kernel with no mixing adds nothing ($0.947 \to 0.947$) and the whole gain is the oblique mixing ($\to 0.965$). The interactions California needs are axis-aligned and reached by the radial kernel without mixing (Prop. 2); those Breast Cancer needs live along learned feature combinations.

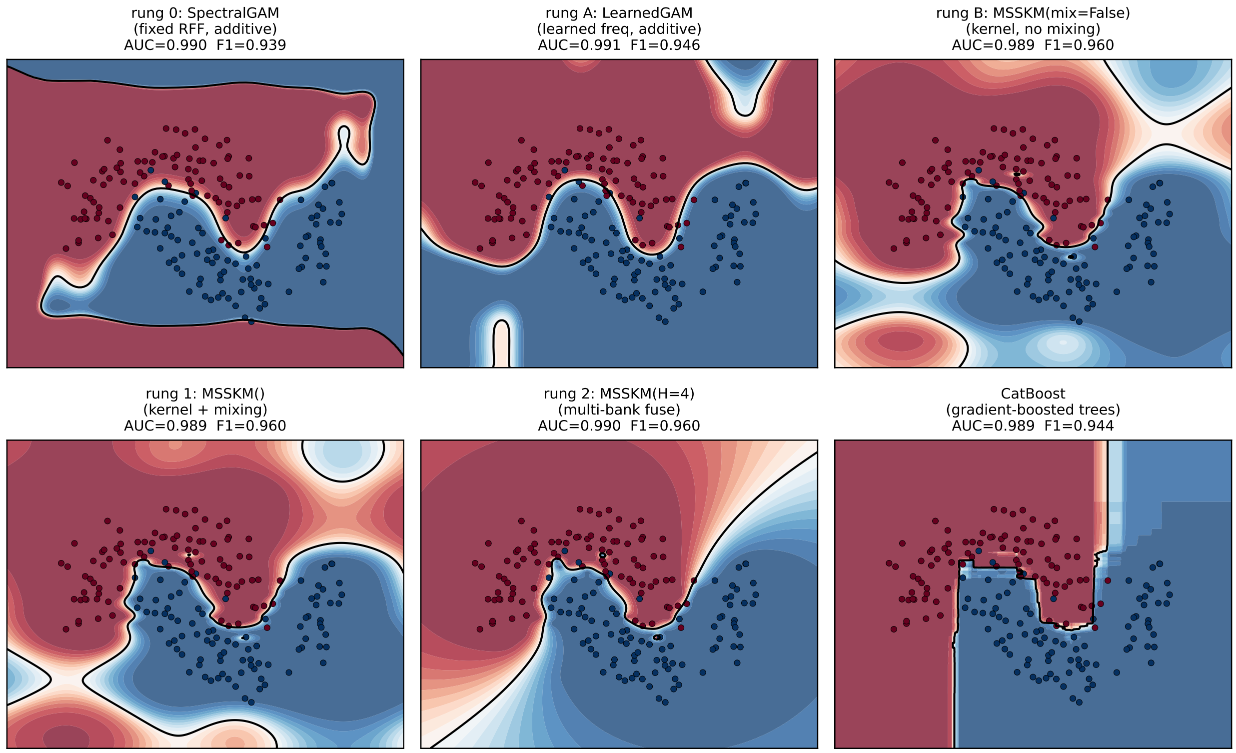

Figure 1: Decision boundaries on `make_moons` (shading is P(class 1), black line the 0.5 boundary, dots the test set). Additive rungs give blocky separable boundaries; the kernel curves them (rung B), mixing makes them oblique (rung 1); CatBoost is an axis-aligned staircase.

Learning the frequencies (rung 0 vs A). On smooth California, learning the frequencies moves test R^2 by < 0.01 ($0.713 \to 0.711$), and on Breast Cancer it is unchanged. This is the flat-spectrum case of Prop. 7: a fixed grid is already a consistent quadrature and the amplitudes carry the information.

Convex fuse versus a linear mixer (Prop. 5). At a matched frequency budget of 256 per feature, one bank with K=256 (a linear mixer over the pooled frequencies) matches one bank with K=64 on California (0.876 vs 0.876) and is worse on Breast Cancer (0.939 vs 0.965, overfitting), while splitting the same 256 frequencies into four convex-fused banks of K=64 reaches 0.885 on California. Pooling frequencies into one kernel does nothing; the gain is the per-bank bandwidth of the convex fuse, exactly Prop. 5.

The $H \times K$ sweep. Table 3 sweeps banks against per-band frequencies, and bears out Prop. 6 and Remark 1. Along each row (fixed H, increasing K) the metric is flat or declines: extra frequencies in one bank do not help and the ungated encoder eventually overfits, sharpest on small Breast Cancer where K=32 at H=8 falls back. Down each column (fixed K, increasing H) the multi-scale regressions improve monotonically (California $0.873 \to 0.885$, Bike-hourly $0.943 \to 0.954$ at K=8), while small Breast Cancer is flat-to-worse, its near-uniform fusion weights signalling no multi-scale structure to exploit. The best corner is moderate K with larger H: capacity comes from banks, not from frequencies per bank.

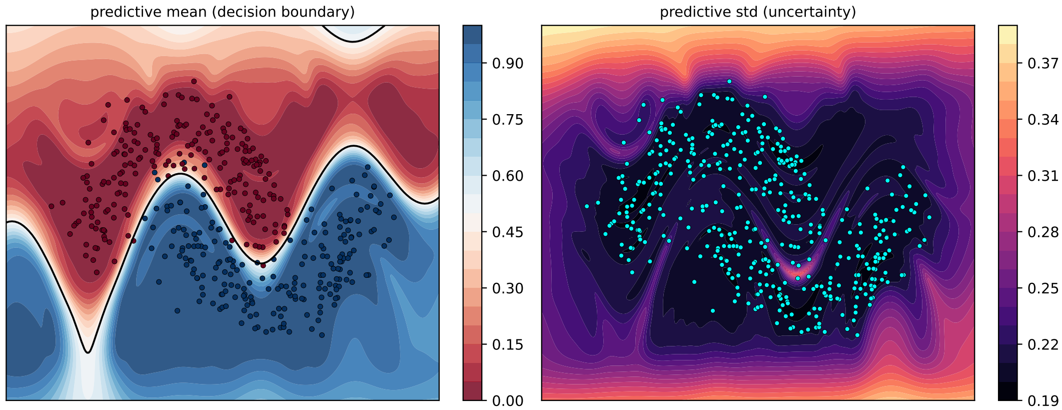

Figure 2: The variational (ELBO) head on `make_moons`, regressing the 0/1 labels. Left: predictive mean with the 0.5 boundary. Right: predictive standard deviation — low over the two crescents where there is data, high in the empty regions. This is the latent predictive spread, not a calibrated Bernoulli posterior; a classification likelihood is future work.

Table 2: The ladder, test metric (R^2 for California, accuracy for Breast Cancer). Each rung lifts one restriction.

rung	model	California R^2	Breast Cancer acc
0	fixed RFF, additive, no kernel	0.713	0.947
A	learned freq, additive, no kernel	0.711	0.947
B	kernel, no mixing (block-diagonal W)	0.873	0.947
1	kernel, full mixing	0.876	0.965
2	multi-bank convex fuse (H=4)	0.885	0.956

Interpretability from the learned kernel. The kernel is learned, so its parameters are the explanation, read off with no surrogate or permutation (Figure 3). *Relevance.* The scale s_j in (2) is the inverse length scale of feature j: it multiplies x_j inside every $\cos / \sin(2\pi s_j \omega_{j,k} x_j)$, so as $s_j \to 0$ those coordinates flatten to constants and feature j leaves the kernel. Training therefore drives s_j small for uninformative features and large for relevant ones, which is automatic relevance determination [14, 16]. *Importance.* Relevance alone ignores how much spectral energy a feature carries, so we score feature j by the sensitivity of the embedding to it. Differentiating (2) gives $\partial_{x_j} \psi(x)_{j,k} = 2\pi s_j \omega_{j,k} a_{j,k} \, [-\sin, \cos](2\pi s_j \omega_{j,k} x_j)$, whose squared magnitude averages to $(2\pi)^2 s_j^2 \omega_{j,k}^2 a_{j,k}^2$ over x_j; summing over the K frequencies and the convex bank weights w_h gives the importance

$$I_j \;\propto\; s_j^2 \sum_h w_h \sum_k a_{h,j,k}^2 \, \omega_{h,j,k}^2, \tag{12}$$

the bank-weighted mean-square gradient of the embedding in x_j — a derivative-based global sensitivity measure [22], comparable across features because the inputs are standardized. On California the importance concentrates on the geographic coordinates, whose effect on price is sharp and local; a strong but smooth feature such as median income carries high relevance yet low gradient energy. The off-diagonal blocks of the encoder metric $M = W^\top W$ (Prop. 2) expose which pairs the model couples — here the household-structure features (average occupancy with average bedrooms and rooms) and the latitude–longitude pair. This is the metric interaction of Prop. 2 made visible; with

Table 3: Test metric over H (banks) $\times$ K (frequencies per feature): R^2 for California and Bike-hourly, accuracy for Breast Cancer. **Bold** is the configuration selected on the validation fold and evaluated once on test. Generated by `benchmarks/sweep_hk.py`.

Dataset	H	K=8	K=16	K=32
California (R^2)	1	0.873	0.875	0.875
	2	0.881	0.880	0.882
	4	**0.883**	0.883	0.882
	8	0.885	0.884	0.884
Breast Cancer (acc)	1	0.956	0.974	0.956
	2	0.965	0.965	**0.982**
	4	0.956	0.956	0.965
	8	0.947	0.974	0.965
Bike-hourly (R^2)	1	0.943	0.942	0.944
	2	0.951	0.952	0.946
	4	0.953	0.953	0.953
	8	0.954	**0.956**	0.953

W block-diagonal (rung B) it is identically zero.

10 Related work

Spectral-mixture [25], sparse-spectrum [13] and à-la-carte [27] kernels learn d-dimensional frequencies that couple all coordinates; (2) is deliberately the opposite, additive before any mixing, with interactions introduced by an inspectable operator. Non-stationary spectral kernels [21] relax stationarity. Deep kernel learning [26] warps inputs by a deep network and can collapse [17]; our warp is shallow and linear by design, which restricts and exposes the collapses rather than precluding them. The per-coordinate Fourier embedding is the periodic numerical embedding of tabular deep learning [7, 28], here given a spectral reading. Convex kernel fusion has the exact decomposition and oracle guarantees of the sum RKHS [1, 2, 23]. Gradient-boosted trees remain the tabular default [3, 8, 11, 18].

11 Discussion

The ladder turns one model into a sequence of controlled questions. Interactions come from the kernel, not the mixer; the mixer only rotates them. Multiple banks beat more frequencies only because the convex fuse keeps a bandwidth per scale; a linear mixer over banks is one wider bank. The per-band frequency count is the overfitting axis and the bank count the multi-scale axis. These are not asymptotic claims but statements read off real data by lifting one restriction at a time. Three directions remain: a broader multi-seed benchmark against gradient boosting, fusion weights chosen by an unbiased risk estimate to inherit the oracle guarantee and a low-rank input projection before the per-coordinate map for genuinely multivariate frequencies.

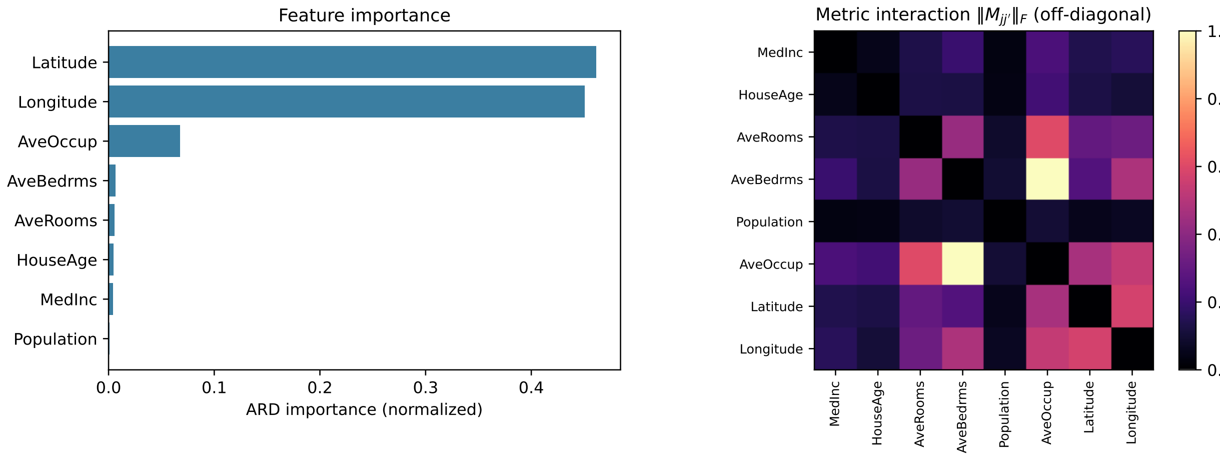

Figure 3: ARD interpretability on California Housing, read directly from the fitted kernel (H=4, K=8). Left: per-feature importance I_j of (12). Right: metric interaction $\|M_{jj'}\|_F$ between feature pairs, diagonal removed.

References

[1] Nachman Aronszajn. Theory of reproducing kernels. *Transactions of the American Mathematical Society*, 68(3):337–404, 1950.

[2] Alain Berlinet and Christine Thomas-Agnan. *Reproducing Kernel Hilbert Spaces in Probability and Statistics*. Springer, 2004.

[3] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *KDD*, 2016.

[4] David Duvenaud, Hannes Nickisch, and Carl E. Rasmussen. Additive gaussian processes. In *NeurIPS*, 2011.

[5] Yarin Gal and Richard Turner. Improving the gaussian process sparse spectrum approximation by representing uncertainty in frequency inputs. In *ICML*, 2015.

[6] Gene H. Golub, Michael Heath, and Grace Wahba. Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2):215–223, 1979.

[7] Yury Gorishniy, Ivan Rubachev, and Artem Babenko. On embeddings for numerical features in tabular deep learning. In *NeurIPS*, 2022.

[8] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *NeurIPS Datasets and Benchmarks*, 2022.

[9] Trevor J. Hastie and Robert J. Tibshirani. *Generalized Additive Models*. Chapman and Hall, 1990.

[10] James Hensman, Nicolò Fusi, and Neil D. Lawrence. Gaussian processes for big data. In *UAI*, 2013.

[11] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *NeurIPS*, 2017.

[12] Cornelius Lanczos. An iteration method for the solution of the eigenvalue problem of linear differential and integral operators. *Journal of Research of the National Bureau of Standards*, 45(4):255–282, 1950.

[13] Miguel Lázaro-Gredilla, Joaquin Quiñonero Candela, Carl E. Rasmussen, and Aníbal R. Figueiras-Vidal. Sparse spectrum gaussian process regression. *Journal of Machine Learning Research*, 11:1865–1881, 2010.

[14] David J. C. MacKay. Bayesian methods for backpropagation networks. In *Models of Neural Networks III*, pages 211–254. Springer, 1996.

[15] Charles A. Micchelli, Yuesheng Xu, and Haizhang Zhang. Universal kernels. *Journal of Machine Learning Research*, 7:2651–2667, 2006.

[16] Radford M. Neal. *Bayesian Learning for Neural Networks*, volume 118 of *Lecture Notes in Statistics*. Springer, 1996.

[17] Sebastian W. Ober, Carl E. Rasmussen, and Mark van der Wilk. The promises and pitfalls of deep kernel learning. In *UAI*, 2021.

[18] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *NeurIPS*, 2018.

[19] Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *NeurIPS*, 2007.

[20] Carl E. Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning.* MIT Press, 2006.

[21] Sami Remes, Markus Heinonen, and Samuel Kaski. Non-stationary spectral kernels. In *NeurIPS*, 2017.

[22] Ilya M. Sobol and Sergei Kucherenko. Derivative based global sensitivity measures and their link with global sensitivity indices. *Mathematics and Computers in Simulation*, 79(10):3009–3017, 2009.

[23] Charles M. Stein. Estimation of the mean of a multivariate normal distribution. *Annals of Statistics*, 9(6):1135–1151, 1981.

[24] Michalis K. Titsias. Variational learning of inducing variables in sparse gaussian processes. In *AISTATS*, 2009.

[25] Andrew Gordon Wilson and Ryan P. Adams. Gaussian process kernels for pattern discovery and extrapolation. In *ICML*, 2013.

[26] Andrew Gordon Wilson, Zhiting Hu, Ruslan Salakhutdinov, and Eric P. Xing. Deep kernel learning. In *AISTATS*, 2016.

[27] Zichao Yang, Alexander J. Smola, Le Song, and Andrew Gordon Wilson. à la carte — learning fast kernels. In *AISTATS*, 2015.

[28] Han-Jia Ye, Huai-Hong Yin, and De-Chuan Zhan. Revisiting nearest neighbor for tabular data: A deep tabular baseline two decades later. *arXiv:2407.03257*, 2024.